

A lossless and fully parallel spintronic compute-in-memory macro for artificial intelligence chips

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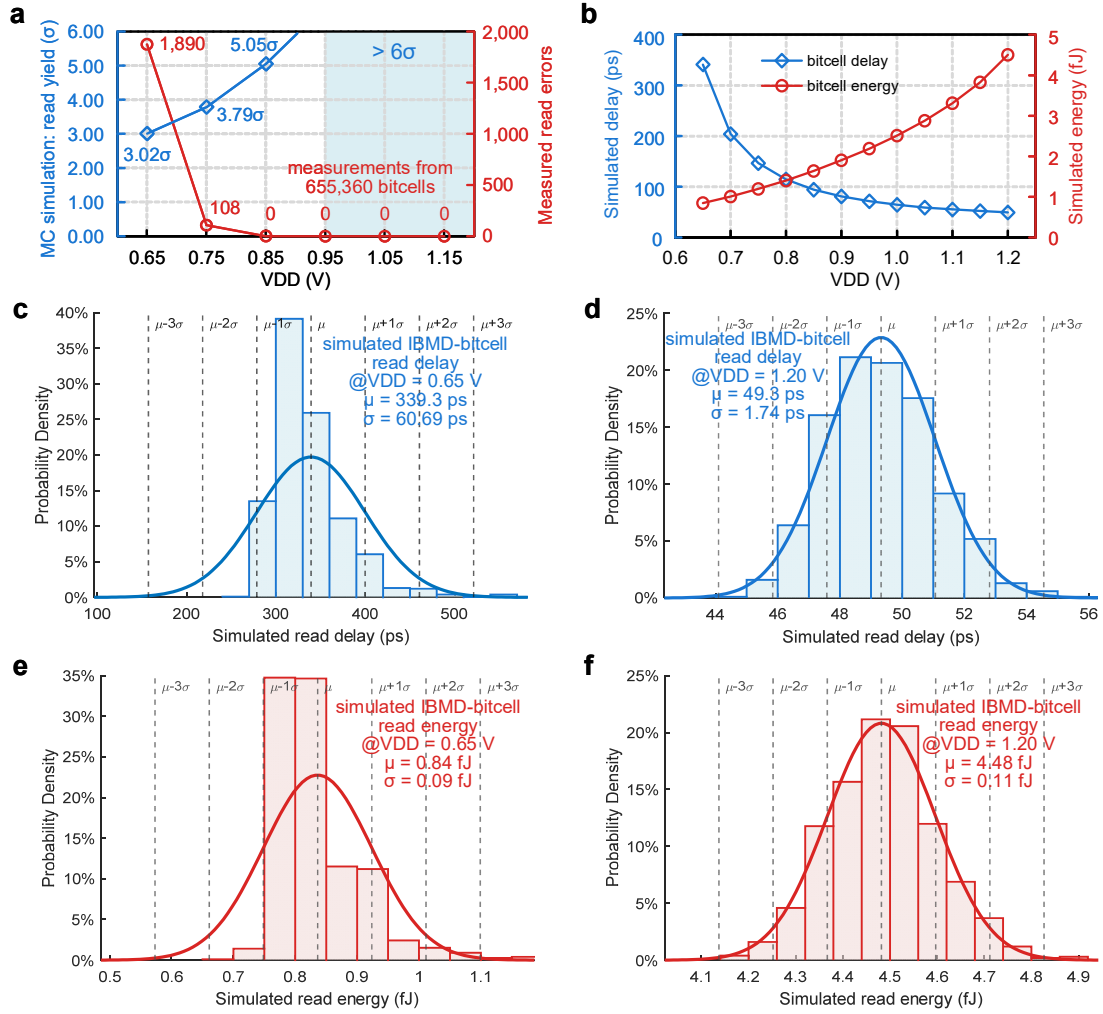
Supplementary Information

Supplementary Table 1

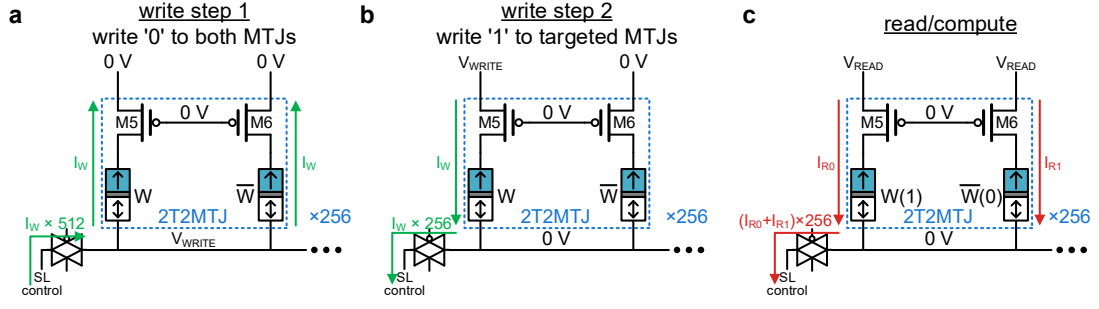
Supplementary Table 1. Comparison of our work with recent spintronic-CMOS and approximate computing works^{1–12}.

References	Memory type	Key focus	Advantages	Fully parallel CIM	Lossless CIM	Bitcell level design	Macro level design	Algorithm level optimization	Chip fabrication
[1] Yang et al., 2022	SOT-MRAM	Hybrid spintronic-CMOS binary logic-in-memory	Energy-efficient CIM operations; reduced data movement	No	Yes	Yes	No	No	No
[2] Fathollahi et al., 2025	SHE-STT-MRAM	Hybrid spintronic-CMOS ternary logic-in-memory	Significant latency and energy savings	No	Yes	Yes	Yes	No	No
[3] Chiu et al., 2023	STT-MRAM	Secure spintronic CIM macro for AI edge devices	High energy efficiency CIM; secure access with PUF	No	Yes	Yes	Yes	No	Yes
[4] Sayadi et al., 2025	N/A	Balancing precision & efficiency in error-resilient applications	Built-in error compensation improves energy efficiency	No	No	No	Yes	Yes	No
[5] Yang et al., 2025	SRAM	Time-domain CIM processor for arbitrary quantized DNNs	Energy-efficient convolution for low-bit-width DNNs	Yes	No	Yes	Yes	No	Yes
[6] Jahannia et al., 2021	Non-Volatile Memory (NVM)	Energy-efficient memory design for edge devices	Clock-frequency and data-lifetime-aware optimization	No	No	No	No	Yes	No
[7] Rezaei et al., 2025	STT-MRAM	Spintronic-based associative memory with CNTFETs	Optimized for energy and area; suitable for neural networks	No	Yes	Yes	Yes	Yes	No
[8] Yu et al., 2024	SRAM, DRAM, NVM	Comprehensive review of CIM technologies for neural networks	Highlights trends and challenges from device design to application	No	No	No	No	No	No
[9] Angizi et al., 2024	SOT-MRAM	Hybrid spin-CMOS polymorphic logic gate	Reconfigurable Boolean logic functions for CIM	No	Yes	Yes	Yes	No	No
[10] Tong et al., 2024	VGSOT-MRAM	Balanced symmetry ternary fully digital in-MRAM computing	Ternary MAC for digital CIM with high energy efficiency	Yes	Yes	Yes	Yes	No	No
[11] Mazaheri et al. 2024	STT-MRAM	Ternary precharge sense amplifier-based MRAM	Highly improved read and write energy and delay with scalability	No	Yes	Yes	Yes	No	No
[12] Rajaei et al. 2020	SHE-STT-MRAM	approximate full-adder based on MTJ	Non-volatile, low area, and low energy consumption for noisy image filter	No	No	Yes	No	No	No
This work: nvDCIM	STT-MRAM	Precision-reconfigurable hybrid spintronic-CMOS CIM macro for AI tasks	Fully digital, lossless computations; precision reconfigurability; High energy efficiency	Yes	Yes	Yes	Yes	Yes	Yes

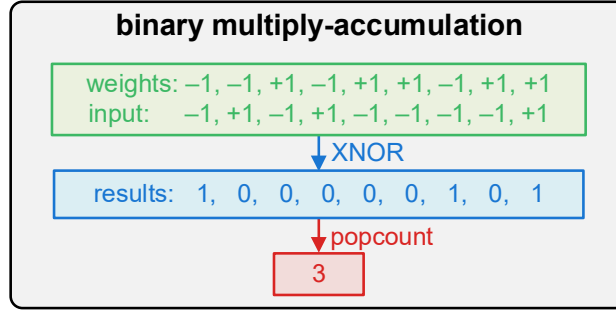
Supplementary Fig. 1–4



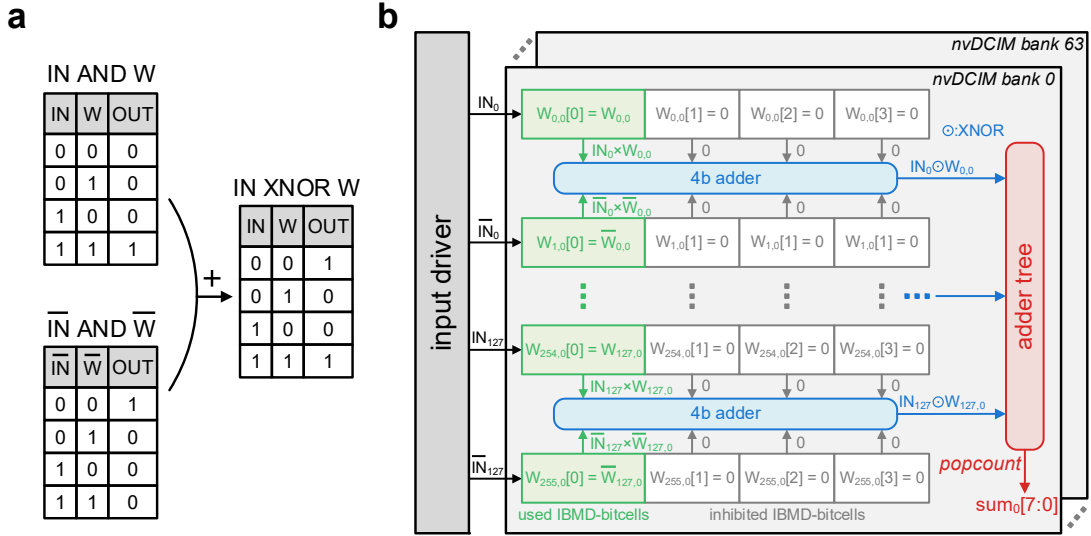
Supplementary Fig. 1. Monte Carlo analysis of the IBMD bitcell. (a) Measured read errors and Monte Carlo simulation results for read yield across supply voltage (VDD) range of 0.65 V–1.15 V. (b) Simulated IBMD-bitcell read delay and energy consumption across VDD range of 0.65 V–1.20 V. (c) Probability distribution of simulated read delay at $VDD = 0.65$ V. (d) Probability distribution of simulated read delay at $VDD = 1.20$ V. (e) Probability distribution of simulated read energy at $VDD = 0.65$ V. (f) Probability distribution of simulated read energy at $VDD = 1.20$ V.



Supplementary Fig. 2. Current flow of IBMD-bitcells. (a) Write operation step 1. (b) Write operation step 2, (c) Read/compute operation.



Supplementary Fig. 3. Binary multiply-accumulation in BNN inference.



Supplementary Fig. 4. Applying proposed nvDCIM macro to BNN. (a) Illustration of XNOR operation based on the proposed IBMD-bitcell, which conducts AND logic. (b) Mapping binary MAC on the proposed nvDCIM macro.

Supplementary Note 1–5

Supplementary Note 1: Comparison with recent hybrid spintronic-CMOS computing and approximate computing

To provide a clear and comprehensive comparison with recent hybrid spintronic-CMOS computing and approximate computing techniques^{1–12}, we have summarized the key focus, advantages, and metrics of the cited works compared to our proposed nvDCIM macro in **Supplementary Table 1**. This comparison helps elucidate the distinctive contributions of our work. Below, we highlight the following key distinctions of our work:

(1) Fully Parallel and Lossless Digital CIM: Many prior works utilize approximate computing, logic-in-memory (LiM), or serial approaches^{1–4,9,11,12}, which often sacrifice accuracy and parallelism. In contrast, our nvDCIM macro performs fully digital, lossless MVM computations with software-equivalent accuracy (post-quantization), a critical requirement for precision-sensitive applications such as PINNs.

(2) Precision Reconfigurability: Existing architectures^{1,2,9,11,12} typically offer fixed or limited precision. Our nvDCIM macro supports precision reconfigurability across 4-bit, 8-bit, 12-bit, and 16-bit modes. This enables dynamic adaptation to various AI workloads, from low-precision CNN to high-precision PINN, enhancing system scalability and versatility.

(3) Algorithm-Level Optimizations: Most prior studies^{1–3,5,8–12} focus on bitcell-level design and macro-level integration, with limited investigation into algorithm-level design. In contrast, our research achieves algorithm-level optimizations based on macro-level design, improving energy efficiency with negligible accuracy degradation.

(4) Chip Fabrication: While many cited studies^{1,2,4,6–12} remain at the simulation level, our nvDCIM macro has been successfully fabricated and experimentally validated, demonstrating its practical feasibility.

As summarized in Table R2-1, while the referenced works contribute significantly to specific areas, they lack a comprehensive solution to key challenges. These challenges include fully parallel, lossless computations, precision reconfigurability, algorithm-level optimizations, and validation through chip fabrication. Our nvDCIM macro bridges this gap and offers a scalable, energy-efficient, and experimentally validated solution for hybrid spintronic-CMOS computing systems aimed at advanced AI

applications.

Supplementary Note 2: nvDCIM macro variations evaluation

When scaling to larger systems or moving toward mass production, quantifying and characterizing these variations is crucial for ensuring the robustness and reliability of our nvDCIM macros in practical applications. We have conducted a comprehensive evaluation of variations at three levels:

(1) Device Level:

Variations in the STT-MRAM properties were experimentally characterized using commercially fabricated wafers. Key parameters, including state-switching voltages

($V_{P \rightarrow AP}$, $V_{AP \rightarrow P}$), resistance (R_{AP} , R_P), and tunnel magneto-resistance (TMR), and their variations were measured. The results are shown in the Methods (STT-MRAM device information) and Extended Data Fig. 1 (STT-MRAM device).

(2) Circuit Level:

At the circuit level, extensive post-layout Monte Carlo simulations were conducted to evaluate the impact of both STT-MRAM device variations and CMOS transistor variations on the performance of the IBMD-bitcell, including its yield, read delay, and energy consumption. The results in Supplementary Fig. 1a indicate that the IBMD-bitcell achieves

a simulated yield of 3.02σ ($\sim 99.75\%$) at 0.65 V, with measurements across 655,360 bitcells revealing 1,890 read errors. At higher V_{DD} (> 0.95 V), the yield improves to $> 6\sigma$. The trade-off between read delay and energy is depicted in Supplementary Fig. 1b. With increasing V_{DD} , the read delay decreases while the read energy increases. At 0.65 V, the simulated read delay is 339.3 ps ($\sigma = 60.69$ ps, Coefficient of Variation (CV) = 17.9%), as shown in Supplementary Fig. 1c, while it reduces to 49.3 ps ($\sigma = 1.74$ ps, CV = 3.5%) at 1.20 V, as shown in Supplementary Fig. 1d. Similarly, the simulated read energy increases from 0.84 fJ ($\sigma = 0.09$ fJ, CV = 10.7%) at 0.65 V, as shown in Supplementary Fig. 1e, to 4.48 fJ ($\sigma = 0.11$ fJ, CV = 2.5%) at 1.20 V, as shown in Supplementary Fig. 1f.

Circuit-level post-layout Monte Carlo simulations demonstrate that the IBMD-bitcell achieves a high yield ($> 6\sigma$ at $V_{DD} > 0.95$ V) with stable read energy and delay characteristics under both low and high supply voltage conditions. This performance highlights the robustness and suitability of IBMD-bitcell for integration into larger

systems.

(3) Chip Level:

To evaluate chip-level variations, we measured the performance and energy efficiency of the nvCIM macros across 24 chips from 12 different shots on the same wafer, with two chips selected per shot. The results are shown in the Methods (Experimental platform and measurement result), and Extended Data Fig. 5 (Measured shmoo plot, energy efficiency, and test results across 24 nvDCIM chips).

To summarize, the device-, circuit-, and chip-level variation evaluations presented above provide a comprehensive understanding of the robustness and reliability of the proposed STT-MRAM nvDCIM macros. They demonstrate that the proposed nvDCIM design exhibits stable performance across different levels of integration, making it a promising candidate for future deployment in larger, more complex systems and mass production.

Supplementary Note 3: Impact of STT-MRAM-based nvDCIM macro on secure AI edge

Security is a critical concern for AI edge devices, especially those deployed in public or IoT environments where they are vulnerable to malicious attacks and unauthorized access to sensitive on-chip data^{13,14}.

While the security aspects of nvDCIM architectures require further exploration, the proposed STT-MRAM-based nvDCIM has inherent advantages for secure AI edge devices. Our proposed STT-MRAM-based nvDCIM architecture offers several inherent advantages that make it a promising candidate for secure AI edge devices:

(1) Intrinsic Security Features of STT-MRAM: The STT-MRAM technology used in our nvDCIM macro provides intrinsic security advantages. For instance, the non-volatility and radiation immunity¹⁵ of MTJs enhance tamper resistance. In addition, the stochastic switching mechanism¹⁶ and process variations of MTJs have been widely used in designing physical unclonable functions (PUFs)^{17–21} and true random number generators (TRNGs)^{22–26}, enabling secure authentication and encryption key generation. For instance, Chiu *et al.*³ reported a STT-MRAM-based CIM macros supporting secure access and authentication through MTJ-based PUFs, which produce highly random and reliable outputs critical for edge security.

(2) Symmetrical IBMD-Bitcell for Side-Channel Attack Resistance: The

IBMD-bitcell in our nvDCIM macro features a symmetrical design with two MTJs storing complementary data. This ensures that current flow and power consumption during read, write, and compute phases are data-independent, maintaining a uniform power profile, as illustrated in Supplementary Fig. 2. As demonstrated in prior studies^{27,28}, such symmetry enhances resilience against side-channel attacks, including differential power analysis (DPA) and correlation power analysis (CPA), by masking data-dependent power variations.

(3) Potential of IBMD-bitcell for PUF and TRNG: The MTJ-based PUF^{17–21} and TRNG^{22–26} leverage the random switching mechanism¹⁶ and process variations of MTJs, typically utilizing sense amplifiers (SAs) or comparators to compare MTJ resistances and generate random outputs²⁹. This principle is similar to the read operation of our IBMD-bitcell, which demonstrates strong potential for PUF and TRNG applications. The IBMD-bitcell can enable unique, repeatable responses for PUFs and high-entropy random number generation for TRNGs. While these applications are beyond the scope of this work, they represent promising future directions.

(4) Energy efficiency, lossless, fully-parallel MVM for AI Edge Devices: AI edge devices require energy-efficient computation to meet tight power budgets, often relying on battery-powered systems. Our nvDCIM macro achieves energy efficiency up to 7.02–112.3 tera-operations per second per watt (TOPS/W), making it highly suitable for energy-constrained edge devices. The toggle-rate-aware training scheme introduced in this work further improves the energy efficiency of AI applications, which reduces energy consumption by up to 109% in ResNet-20 implementations. Furthermore, the nvDCIM macro supports flexible precision configurations (4-bit, 8-bit, 12-bit, and 16-bit) to enable lossless MVMs across various AI workloads, such as ResNet-20 and PINNs, without compromising inference accuracy.

Supplementary Note 4: nvDCIM macro for BNN applications

While our current evaluation focuses on high-precision applications such as Convolutional Neural Networks (CNNs) and Physics-Informed Neural Networks (PINNs), extending the macro to BNNs represents an exciting direction for future work. BNNs are particularly attractive for their ability to reduce model complexity³⁰ by aggressively quantizing parameters to 1-bit precision while maintaining reasonable accuracy^{31,32}. The primary operation in BNN inference is the binary multiply-

accumulation (MAC) operation, which can be expressed as:

$$MAC(IN, W) = \text{popcount}(XNOR(IN, W)) \quad (1)$$

Here, IN and W represent binary-encoded inputs and weights³⁰ (“+1” or “−1”). The operation involves performing bitwise XNOR operations between the inputs and weights, followed by a *popcount* operation to sum the XNOR outputs, as shown in Supplementary Fig. 3. To simplify hardware implementation, this encoding can be transformed into “0” and “1”.

To assess the feasibility of adapting our nvDCIM macro for BNNs, we carefully analyzed the implementation of binary MAC operations and identified several trade-offs in performance, storage density, and energy efficiency. Below, we outline our approach and the associated challenges:

(1) XNOR Logic Implementation:

As the IBMD bitcell in our macro is designed to perform the AND operation, the XNOR operation must be implemented indirectly. Specifically:

$$XNOR(IN, W) = (IN \cdot W) + (\overline{IN} \cdot \overline{W}) \quad (2)$$

This requires two IBMD bitcells: one for computing $(IN \cdot W)$ and the other for computing $(\overline{IN} \cdot \overline{W})$. The results are then summed using an adder (or an OR gate) (Supplementary Fig. 4a).

(2) Mapping Binary MAC Operations:

The binary MAC operations are mapped onto the nvDCIM macro as shown in Supplementary Fig. 4b. Each binary weight W and its complement $\overline{W}$ are stored in two adjacent bitcells. During operation, the input IN and its complement $\overline{IN}$ are applied to these bitcells, and their outputs are summed using an adder to get the XNOR results. The *popcount* functionality is implemented by accumulating these results through the remaining adder tree.

(3) Challenges and Trade-offs:

1. **Storage density:** Each binary weight requires two bitcells, and only 1/4 of the bitcells in a bank store effective value, with the remaining 3/4 inhibited (set to 0) to ensure correct functionality. This significantly reduces storage density and throughput.
2. **Performance:** Binary MAC operations can be executed in a single clock cycle, while 4-bit precision MAC operations require four clock cycles. Additionally, binary MAC operations benefit from a shorter critical path, potentially enabling

a higher clock frequency. However, since each weight requires two bitcells and only 1/4 of the bitcells are actively used in the macro, the overall throughput is estimated to decrease to $\sim 1/2$ compared to 4-bit precision MAC at the same clock frequency.

- 3. Energy Efficiency:** The reduced throughput directly impacts energy efficiency, which is expected to decrease significantly compared to 4-bit precision tasks such as CNNs.

Despite these challenges, implementing BNNs on our macro remains promising and demonstrates its potential for improving its reconfigurability to support both high- and low-precision applications.

Supplementary Note 5: Challenge analysis of STT-MRAM

While the integration of STT-MRAM offers several key advantages, including non-volatility, high endurance, and CMOS compatibility, its widespread commercialization is constrained by several challenges. Below, we discuss the key scalability issues, area overhead, and inherent limitations of the technology.

Scalability Challenges: While individual MTJ has demonstrated scalability down to a diameter of ~ 2.0 nm with 10-year data retention^{33,34}, achieving high-density MTJ arrays remains challenging due to their unique etching difficulties. Current MTJ fabrication typically relies on ion beam etching (IBE)³⁵, as conventional reactive ion etching (RIE) is not yet effective for MTJ materials, which do not form volatile by-products³⁶. However, IBE introduces issues such as material re-deposition and sidewall damage³⁷, which can lead to electrical shorts or weak points³⁸, and limit the minimum pitch between adjacent MTJs³⁹, thereby constraining array-level scalability toward sub-5 nm technology nodes.

Area Overhead: The standard 1T1M (one-transistor, one-MTJ) cell occupies approximately $30F^2$ (F : minimum lithography feature size), offering a higher density than SRAM (6T: $\sim 140F^2$)⁴⁰. Notably, at the IEDM 2016, Toshiba and SK Hynix demonstrated a 4-Gbit STT-MRAM chip⁴¹ with a cell area of $9F^2$, comparable to DRAM ($6F^2$ – $8F^2$)^{42–44}.

Regarding the CIM applications, our IBMD-bitcell, which supports in-situ multiplication and digitization for lossless and fully-parallel MVMs in the digital domain, occupies approximately $1600F^2$. This area is comparable to other nvCIM

implementations, such as MRAM-based⁴⁵ design (2T2M: $\sim 1200F^2$) and PCM-based⁴⁶ design (8T4R: $\sim 25,000F^2$), and could be further optimized in the future.

Limitations: (1) Manufacturing complexity: The MTJ structure incorporates some materials (Co, Fe, Mg, etc.) that are less commonly used in the standard Si-based CMOS process, which demands new equipment and process adjustments to the existing fabrication flow^{47,48}. This increases fabrication cost and reduces throughput. (2) Etching difficulty: As mentioned above, MTJ materials face challenges in etching, due to the noble metal components, such as Pt, Pd, and Ir, which are unable to produce volatile by-products using RIE. Although IBE is typically employed, it suffers from issues such as sidewall damage and material re-deposition^{49,50}. Advanced multi-step etching strategies can mitigate these issues, but further increase process complexity and decrease throughput. (3) Unlimited endurance challenges: Although MTJ devices have demonstrated endurance exceeding 10^{16} cycles⁵¹ ($V_{\text{WRITE}} = 0.65V$, 10 ns pulse) at the device level, achieving the “unlimited endurance” benchmark of SRAM and DRAM, the endurance remains challenging for MRAM commercial chips. For instance, at IEDM 2022, Samsung’s 16 Mb MRAM chip⁵² achieved endurance over 10^{14} cycles at $-25^\circ C$. While this falls short of SRAM/DRAM ($>10^{16}$), it is higher than other NVMs, including PCM (chip⁵³: 10^8 cycles, device⁵⁴: 10^{12} cycles), RRAM (chip⁵⁵: 10^7 cycles, device⁵⁶: 10^{12} cycles), and HZO-based FRAM (chip^{57,58}: 10^{12} cycles, device⁵⁹: 10^{15} cycles).

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